

17/7/2026

Analytical test

Shanya.P.S
192524014

- 1] using the Caesar cipher to encrypt the message
"HELLO" ($K=5$)

Sol

Plain text = HELLO

using formula $E = (P+K) \bmod 26$

$$'H' = (7+5) \bmod 26 = 12 \Rightarrow 'M'$$

$$'E' = (4+5) \bmod 26 = 9 \Rightarrow 'J'$$

$$'L' = (11+5) \bmod 26 = 16 \Rightarrow 'Q'$$

$$'L' = (11+5) \bmod 26 = 16 \Rightarrow 'Q'$$

$$'O' = (14+5) \bmod 26 = 19 \Rightarrow 'T'$$

Cipher text = MJQQT

- 2] Caesar cipher

Plaintext = "meet me after the party"
($K=3$)

Sol

Plaintext = "meet me after the party"

$$K = 3$$

$$E = (P+K) \bmod 26$$

$$m = (12+3) \bmod 26 = 15 = P$$

$$e = (4+3) \bmod 26 = 7 = h$$

$$e = (4+3) \bmod 26 = 7 = h$$

$$t = (19+3) \bmod 26 = 22 = w$$

me
after
the
party

$$m = (12+3) \bmod 26 = 15 = P$$

$$e = (4+3) \bmod 26 = 7 = h$$

$$a = (0+3) \bmod 26 = 3 = d$$

$$\begin{aligned}
 t &= (19 + 3) \bmod 26 = 22 = w \\
 e &= (4 + 3) \bmod 26 = 7 = h \\
 r &= (11 + 3) \bmod 26 = 20 = u \\
 t &= (19 + 3) \bmod 26 = 22 = w \\
 H &= (7 + 3) \bmod 26 = 10 = k \\
 e &= (4 + 3) \bmod 26 = 7 = h \\
 P &= (15 + 3) \bmod 26 = 18 = s \\
 a &= (0 + 3) \bmod 26 = 3 = d \\
 r &= (11 + 3) \bmod 26 = 20 = u \\
 t &= (19 + 3) \bmod 26 = 22 = w \\
 y &= (24 + 3) \bmod 26 = 21 = b
 \end{aligned}$$

CT = "Phhw Ph diwhu wrh squwB"

3] mono alphabetic cipher

PT = "meet me tomorrow in the morning"

SOL

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G
U	V	W	X	Y	Z														
F	E	D	C	A	B														

Cipher text = "NVYG NV GLNLILDP RM
QSV NLIMRMT"

4] Playfair cipher

PT = "MEET me tomorrow"

key = keyword

K	E	Y	W	O
R	D	A	B	C
F	G	H	I	L
M	N	P	Q	R
S	T	U	V	X

K	E	Y	W	O
R	D	A	B	C
F	G	H	I	L
M	N	P	Q	R
T	U	V	X	Z

meetmetomorrow

me et me to mo rx ro wx

KN UK KN KZ SK BT CK BW

Cipher text = "KN UK KN KZ SK BT CK BW"

5] poly alphabetic cipher
Vigenere cipher

PT = "ACTIVE"

Key = "PASCAL"

A	C	T	I	V	E
0	2	19	8	21	4
P	A	S	C	A	L
15	0	18	2	0	11

15 2 37 10 21 15

$15 \bmod 26$ $2 \bmod 26$ $37 \bmod 26$ $10 \bmod 26$ $21 \bmod 26$ $15 \bmod 26$
 15 2 11 10 21 15
 P C L K V P

CT = "PCLKVP"

6] Hill cipher

PT = "HELP"

Key = $\begin{Bmatrix} 3 & 3 \\ 2 & 5 \end{Bmatrix}$

$$PT = \begin{bmatrix} H & E \\ L & D \end{bmatrix} \Rightarrow \begin{bmatrix} 7 & 4 \\ 11 & 3 \end{bmatrix}$$

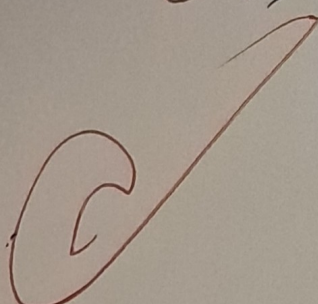
$$H = (PK) \bmod 26$$

$$= \begin{bmatrix} 7 & 4 \\ 11 & 3 \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 21+8 & 21+20 \\ 33+6 & 33+15 \end{bmatrix} = \begin{bmatrix} 29 & 41 \\ 39 & 48 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 29 & 41 \\ 39 & 48 \end{bmatrix} \bmod 26 \Rightarrow \begin{bmatrix} 3 & 15 \\ 13 & 22 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} D & P \\ N & W \end{bmatrix} \Rightarrow CT = "DPNW"$$



2411
 26
 15
 89
 26
 13
 7
 26
 03
 8
 6
 2